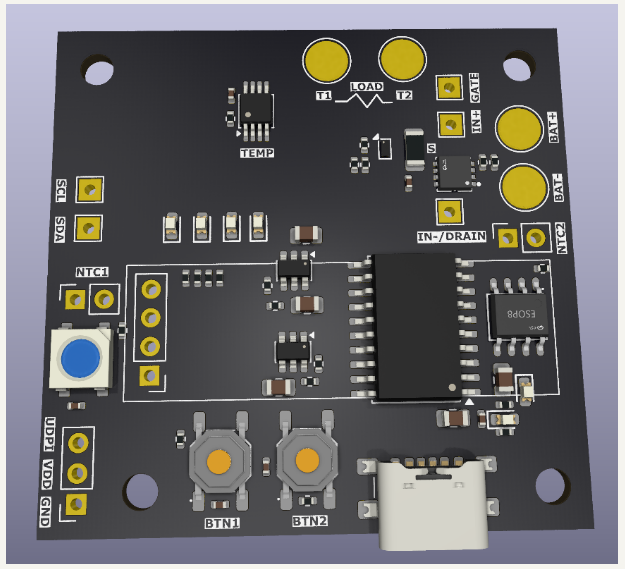
HEATER CONTROLLER

2026-05-18

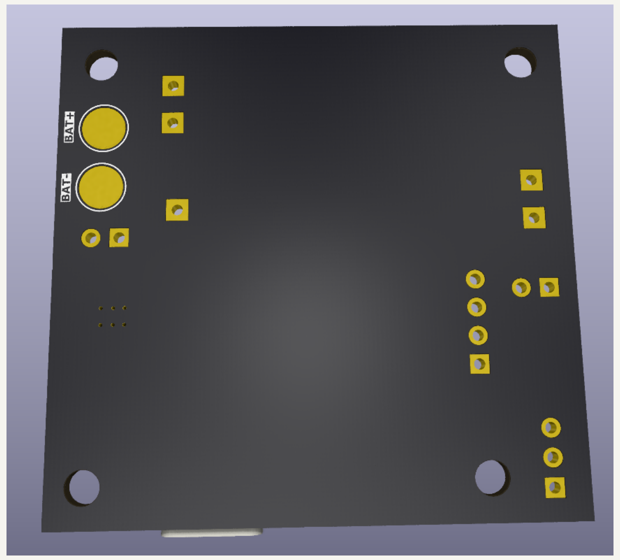
Rev 1

Variant:

TOP VIEW



BOTTOM VIEW



NOTES

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

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Comments:

Company: CAPISTOR TECHNOLOGIES - FZCO

Variant:

Board Name: HEATER CONTROLLER

Project Name: HEATER CONTROLLER

Sheet Title: Cover Page

File Name: HEATER CONTROLLER REV_A.kicad_sch

Designer: SHOAIB MUSTAFA

Date: Last Modified Date

Revision: 1

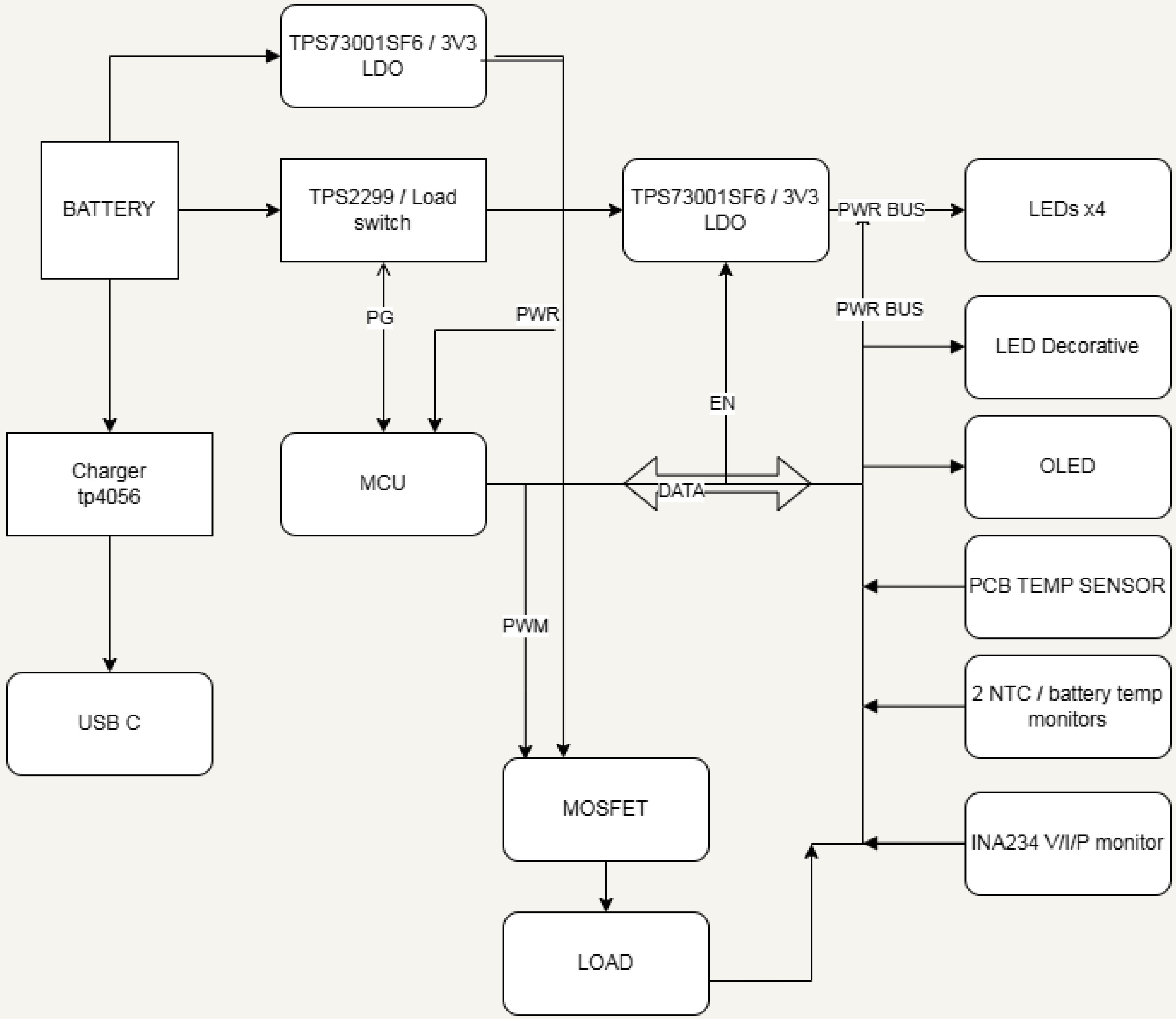
Sheet Path: /

Reviewer:

Size: A3

Sheet: 1 of 7

[2] Block Diagram



Comments:	Company: CAPISTOR TECHNOLOGIES - FZCO		Variant:	
	Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 7

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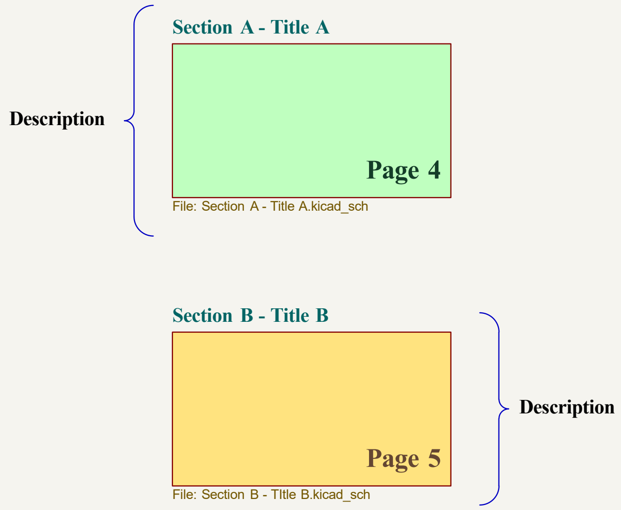
[3] Project Architecture



Comments:	Company: CAPISTOR TECHNOLOGIES - FZCO		Variant:	
	Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 7

Undervoltage protection by MCU mechanism
1- mcu in low power mode 20-60uA
2- All gpios to off state
3- LDO control turned off
4- mosfet gpio off (1uA from mosfet only)
5- INA234 only 2.2uA at standby
2.2 + 1 + 60 = 60.3uA
is the max current required for the mcu to stay on and keep the battery current limited acting as UV cutoff

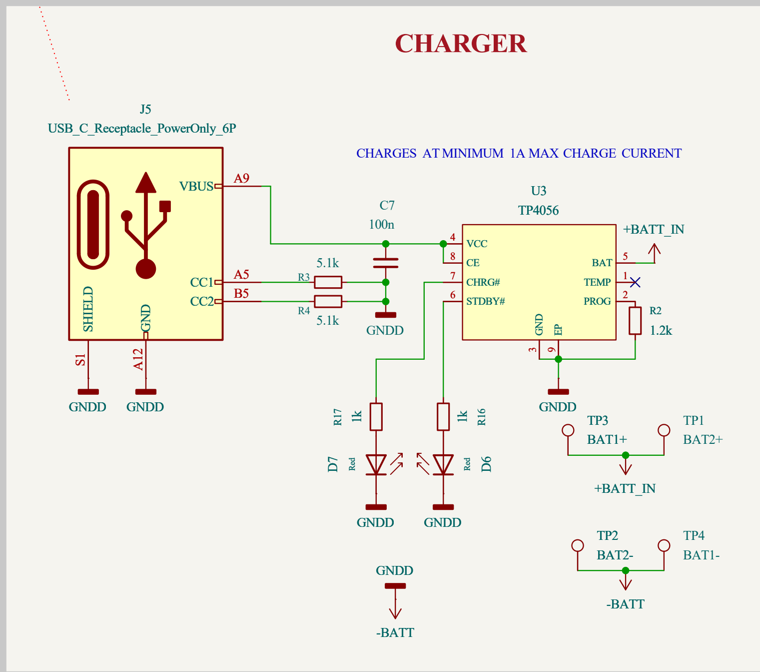
Calculation for CSD MOSFET
at V(gs) 2-4V ~ 2-4nC
at V(gs) 2-4V ~ 8-20mohms R(ds)ON
pulsed gate current rating increases from 10A -> 100A depending on duty cycle
Temperature satisfactory under load as per datasheet
As per calculations switching speed of up to 4.4nS assuming 30+ 330 series resistance and 4nC

LAYOUT NOTE:
MUST BE CLOSE TO THE HEATER PADS TO MONITOR POINT OF FAILURE

SHUNT RESISTOR:
8A max current setting
This value can be changed by changing the shunt resistor
smaller can be selected but will be reduced accuracy

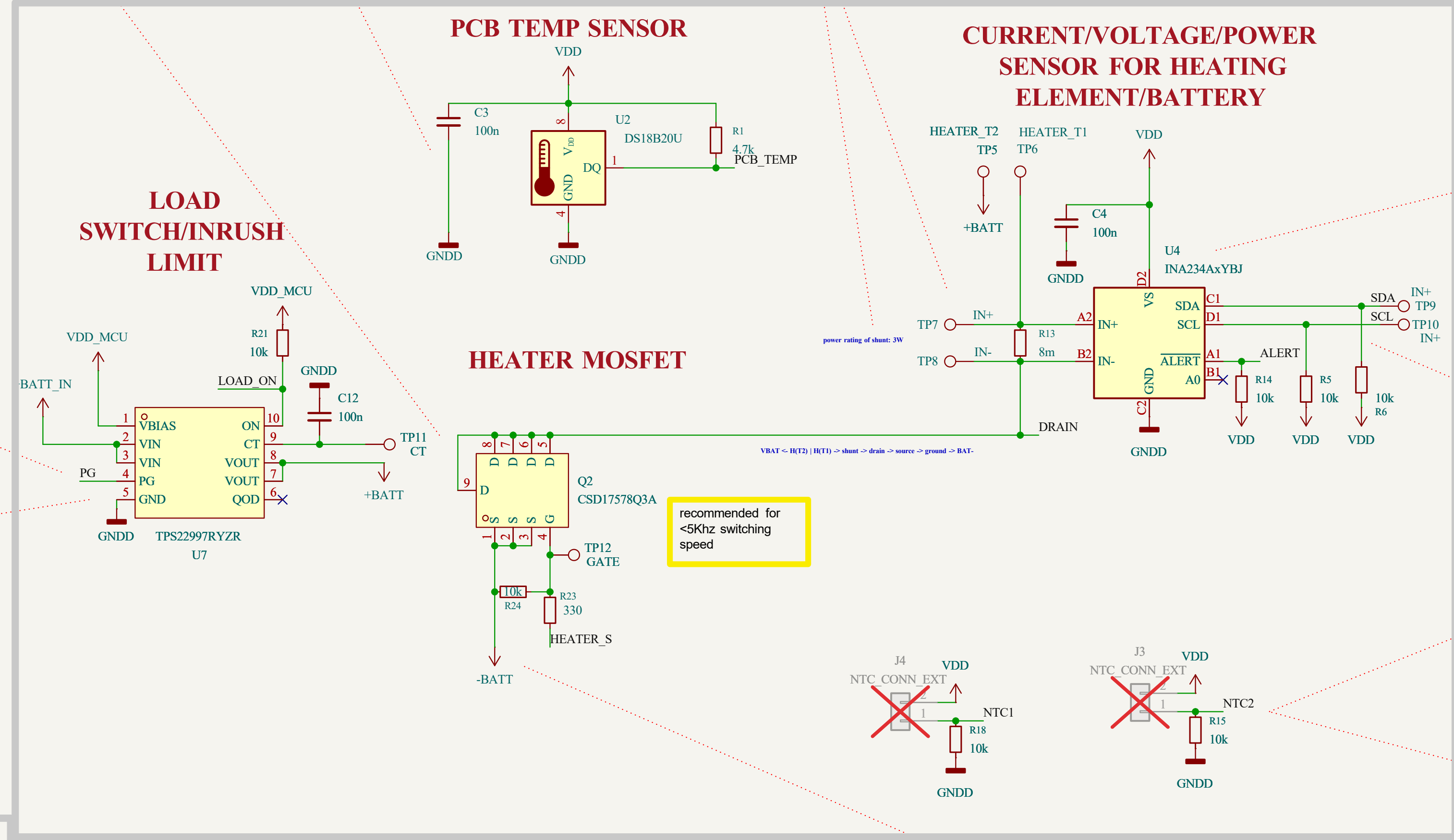
Inrush current capabillity

6pin usbc only charge



Max 10A, soft inrush, configurable, slew rate

VBIAS within VDD_MCU range (checked)
VIN within the VBAT range (checked)
Rise and fall time configurable around 1.1mS
10uA Qs current



recommended for <5Khz switching speed

monitors Voltage rail getting to mcu. i.e; bat -> ldo -> mcu
Also measures the voltage in the load
Measures the current/power in the load
option to set under, over for supply as well as load bus

Refer to setting up startup register mechanism defined in the datasheet for proper operation in firmware

DESIGN NOTE:
VDD will shift when battery voltage is below 3.3V. causing inaccurate ntc readings. below VDD ~ 3.3V firmware clauses have to be added to handle that situation

According to section 18.2 of attiny3216 documentation internal VREF can be select if configured in the mcu which are more stable. It is recommended to use that as opposed to VDD domain

DESIGN NOTE:
Two ntc needed for two batteries.
ASSUMING NTC OF THIS TYPE WITH THE FOLLOWING LOOKUP TABLE
Resistance at 25 degrees C: 10K +- 1%
B25/50 = 3950+- 1%
Thermal time constant <= 15 seconds
Thermistor temperature range -55 °C to 125 °C
28 AWG PVC Wire

Resistance @ 25°C = 10KΩ $B_{25/50}$=3950K

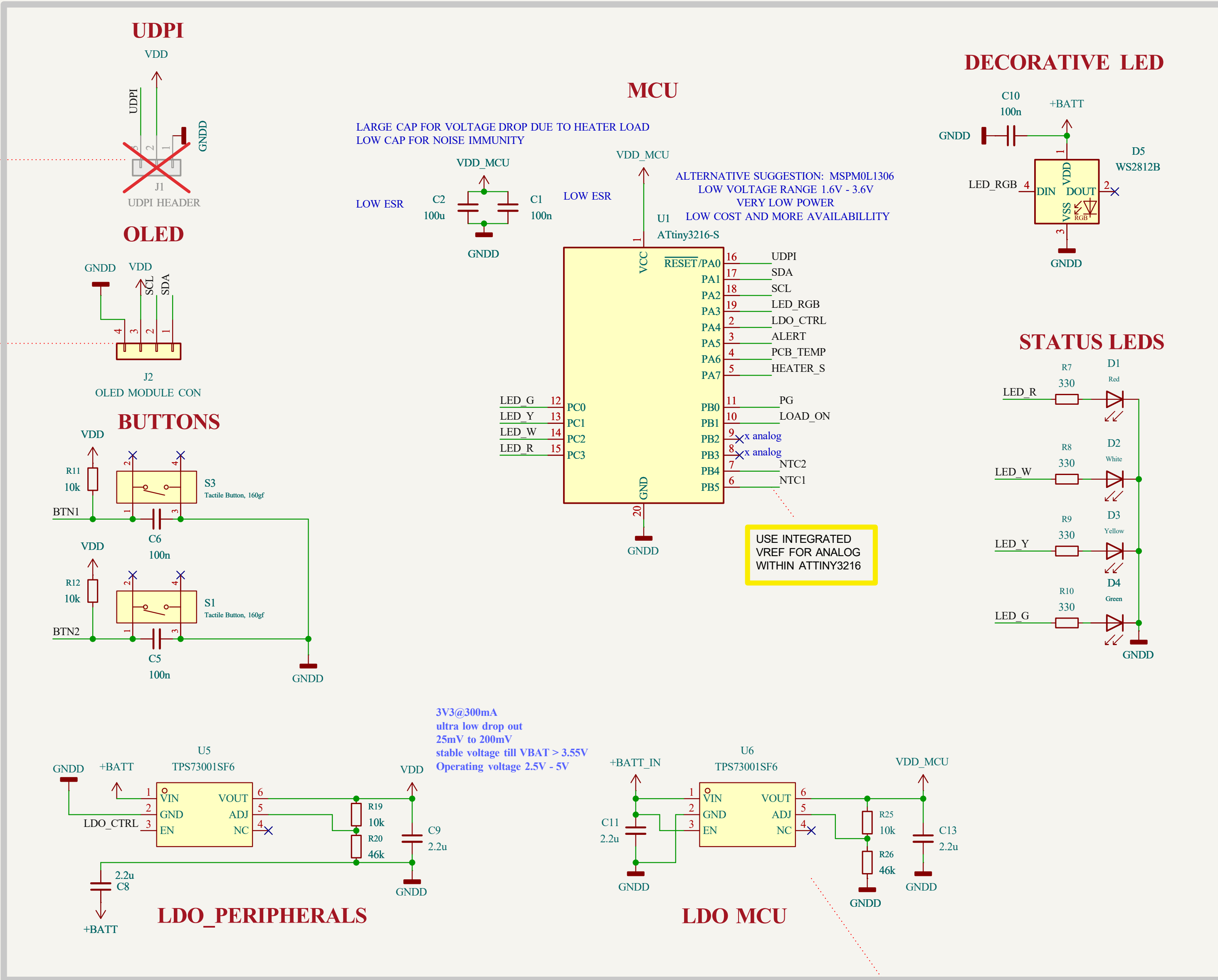
T(°C)	R(KΩ)	T(°C)	R(KΩ)	T(°C)	R(KΩ)	T(°C)	R(KΩ)	T(°C)	R(KΩ)	T(°C)	R(KΩ)
-40	277.2	1	30.25	42	4.915	83	1.128	124	0.3434	165	0.1265
-39	263.6	2	28.82	43	4.723	84	1.093	125	0.3341	166	0.1239
-38	250.1	3	27.45	44	4.539	85	1.059	126	0.3253	167	0.1213
-37	236.8	4	26.16	45	4.363	86	1.027	127	0.3167	168	0.1187
-36	224.0	5	24.94	46	4.195	87	0.9955	128	0.3083	169	0.1163
-35	211.5	6	23.77	47	4.034	88	0.9654	129	0.3002	170	0.1139
-34	199.6	7	22.67	48	3.880	89	0.9363	130	0.2924	171	0.1115
-33	188.1	8	21.62	49	3.733	90	0.9083	131	0.2848	172	0.1092
-32	177.3	9	20.63	50	3.592	91	0.8812	132	0.2774	173	0.1070
-31	167.0	10	19.68	51	3.457	92	0.8550	133	0.2702	174	0.1048
-30	157.2	11	18.78	52	3.328	93	0.8297	134	0.2633	175	0.1027
-29	148.1	12	17.93	53	3.204	94	0.8052	135	0.2565	176	0.1006
-28	139.4	13	17.12	54	3.086	95	0.7816	136	0.2500	177	0.0986
-27	131.3	14	16.35	55	2.972	96	0.7587	137	0.2437	178	0.0966
-26	123.7	15	15.62	56	2.863	97	0.7366	138	0.2375	179	0.0947
-25	116.6	16	14.93	57	2.759	98	0.7152	139	0.2316	180	0.0928
-24	110.0	17	14.26	58	2.659	99	0.6945	140	0.2258	181	0.0909
-23	103.7	18	13.63	59	2.564	100	0.6744	141	0.2202	182	0.0891
-22	97.9	19	13.04	60	2.472	101	0.6558	142	0.2148	183	0.0873
-21	92.50	20	12.47	61	2.384	102	0.6376	143	0.2095	184	0.0856
-20	87.43	21	11.92	62	2.299	103	0.6199	144	0.2044	185	0.0839
-19	82.79	22	11.41	63	2.218	104	0.6026	145	0.1994	186	0.0822
-18	78.44	23	10.91	64	2.141	105	0.5858	146	0.1946	187	0.0806
-17	74.36	24	10.45	65	2.066	106	0.5694	147	0.1900	188	0.0790
-16	70.53	25	10.00	66	1.994	107	0.5535	148	0.1855	189	0.0774
-15	66.92	26	9.575	67	1.926	108	0.5380	149	0.1811	190	0.0759
-14	63.54	27	9.170	68	1.860	109	0.5229	150	0.1769	191	0.0743
-13	60.34	28	8.784	69	1.796	110	0.5083	151	0.1728	192	0.0729
-12	57.33	29	8.416	70	1.735	111	0.4941	152	0.1688	193	0.0714
-11	54.50	30	8.064	71	1.677	112	0.4803	153	0.1650	194	0.0700
-10	51.82	31	7.730	72	1.621	113	0.4669	154	0.1612	195	0.0686
-9	49.28	32	7.410	73	1.567	114	0.4539	155	0.1576	196	0.0672
-8	46.89	33	7.106	74	1.515	115	0.4412	156	0.1541	197	0.0658
-7	44.62	34	6.815	75	1.465	116	0.4290	157	0.1507	198	0.0645
-6	42.48	35	6.538	76	1.417	117	0.4171	158	0.1474	199	0.0631
-5	40.45	36	6.273	77	1.371	118	0.4055	159	0.1441	200	0.0619
-4	38.53	37	6.020	78	1.326	119	0.3944	160	0.1410		
-3	36.70	38	5.778	79	1.284	120	0.3835	161	0.1379		
-2	34.97	39	5.548	80	1.243	121	0.3730	162	0.1350		
-1	33.33	40	5.327	81	1.203	122	0.3628	163	0.1321		
0	31.77	41	5.117	82	1.165	123	0.3530	164	0.1293		

PWR GROUND TO BE SEPERATED WHILE ROUTING

[4]MAIN

LAYOUT NOTE:
INSTALL AT THE EDGE FOR EASY ACCESS FOR POGO PINS AND HEADER PINS

LAYOUT NOTE:
CREATE FOOTPRINT AND ALLOW ADEQUATE SPACE FOR INSTALLING THE OLED MODULE



USE INTEGRATED VREF FOR ANALOG WITHIN ATTINY3216

H1 MountingHole
H3 MountingHole
H4 MountingHole

5.1 Multiplexed Signals

Table 5-1. PORT Function Multiplexing

VQFN 24-Pin	SOIC 20-Pin	Pin Name (1,2)	Other/Special	ADC0	ADC1	PTC(4)	AC0	AC1	AC2	DAC0	USART0	SPI0	TWI0	TCA0	TCBn	TCD0	CCL
23	16	PA0	~~RESET~~/ UPDI	AIN0													LUT0-IN0
24	17	PA1		AIN1							TxD(3)	MOSI	SDA(3)				LUT0-IN1
1	18	PA2	EVOUT0	AIN2							RxD(3)	MISO	SCL(3)				LUT0-IN2
2	19	PA3	EXTCLK	AIN3							XCK(3)	SCK		WO3	TCB1 WO		
3	20	GND															
4	1	V_{DD}															
5	2	PA4		AIN4	AIN0	X0/Y0					XDIR(3)	$\overline{SS}$		WO4		WOA	LUT0-OUT
6	3	PA5	VREFA	AIN5	AIN1	X1/Y1	OUT	AINN0						WO5	TCB0 WO	WOB	
7	4	PA6		AIN6	AIN2	X2/Y2	AINN0	AINP1	AINP0	OUT							
8	5	PA7		AIN7	AIN3	X3/Y3	AINP0	AINP0	AINN0								LUT1-OUT
9		PB7			AIN4			AINN1	AINP3								
10		PB6			AIN5		AINP3		AINN1								
11	6	PB5	CLKOUT	AIN8		X12/Y12	AINP1		AINP2					WO2(3)			
12	7	PB4		AIN9		X13/Y13	AINN1	AINP3						WO1(3)			LUT0-OUT(3)
13	8	PB3	TOSC1					OUT			RxD			WO0(3)			
14	9	PB2	TOSC2, EVOUT1						OUT		TxD			WO2			
15	10	PB1		AIN10		X4/Y4	AINP2				XCK		SDA	WO1			
16	11	PB0		AIN11		X5/Y5		AINP2	AINP1		XDIR		SCL	WO0			
17	12	PC0			AIN6	X6/Y6						SCK(3)			TCB0 WO(3)	WOC	
18	13	PC1			AIN7	X7/Y7						MISO(3)				WOD	LUT1-OUT(3)
19	14	PC2	EVOUT2		AIN8	X8/Y8						MOSI(3)					
20	15	PC3			AIN9	X9/Y9						$\overline{SS}$(3)		WO3(3)			LUT1-IN0
21		PC4			AIN10	X10/Y10								WO4(3)	TCB1 WO(3)		LUT1-IN1
22		PC5			AIN11	X11/Y11								WO5(3)			LUT1-IN2

DESIGN NOTE:
VDD_MUC is always on
VDD ldo can be controlled by mcu

A

Comments:	Company:		Variant:	
OLED MODULE DATASHEET Application note for INA234 J. Pieper ADC investigation	CAPISTOR TECHNOLOGIES - FZCO			
	Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
Sheet Title: Sheet Title A	File Name: Section A - Title A.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1
Sheet Path: /Project Architecture/Section A - Title A/		Reviewer:	Size: A2	Sheet: 4 of 7

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[5] Sheet Title B

Comments: **References:** Flexible I/O worked examples Flexible I/O source configuration	Company: CAPISTOR TECHNOLOGIES - FZCO		Variant:	
	Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
Sheet Title: Sheet Title B	File Name: Section B - TItle B.kicad_sch	Designer: SHOAIB MUSTAFA	Date: Last Modified Date	Revision: 1
Sheet Path: /Project Architecture/Section B - Title B/		Reviewer:	Size: A3	Sheet: 5 of 7

7 Parameter Measurement Information

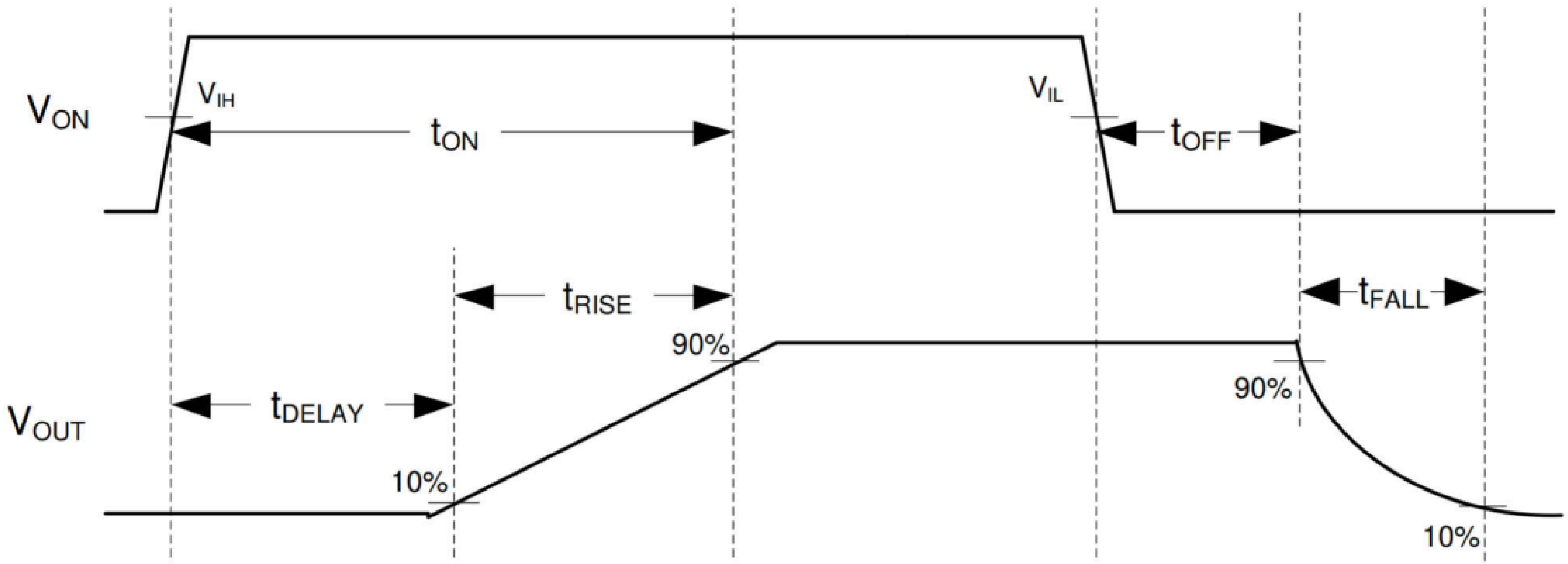

Figure 7-1. TPS22997 Timing Diagram

Comments: REASONING FOR CHOOSING THIS METHOD	Company: CAPISTOR TECHNOLOGIES - FZCO		Variant:	
	Board Name: HEATER CONTROLLER		Project Name: HEATER CONTROLLER	
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Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 6 of 7

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[7] Revision History

DD.MM.YYYY - xxx Revision
Variant: xxx

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